UNIVERSITY of LOUISIANA LAFAYETTE | Kathleen Babineaux Blanco Public Policy Center

POLICY RESEARCH BRIEF
May 2023

POLICY BRIEF
Offshore Wind in the Gulf of Mexico: Natural Resource Revenue Potential

Stephen Barnes, PhD; Matthew Holland, BSBA; Claudia Laurenzano, MS; Anna Osland, PhD; Chenhan Shao, MPP

INTRODUCTION

The Gulf of Mexico has long played a central role in energy production for the United States. Offshore oil and gas drilling in the federal Outer Continental Shelf (OCS) waters of the Gulf of Mexico currently amounts to 15% of the total U.S. crude oil production and 5% of natural gas production.[1] However, the outlook for Gulf of Mexico energy production is changing future. Dwindling reserves in shallow, lower cost lease areas, plus regulatory uncertainty regarding new oil and gas leases in the area have shifted the outlook for future fossil fuel production in the Gulf of Mexico. Simultaneously, the federal government has announced plans to deploy 30 GW of offshore wind energy across the U.S. by 2030.[2] The development of wind energy in the Gulf of Mexico has the potential to leverage Louisiana's existing workforce and strengths in offshore development, diversify the region's energy mix, reduce greenhouse gas emissions, and provide a new source of public revenue.

Federal revenues from oil and gas leases in the Gulf of Mexico are allocated to nearby state and local governments through a structure established by the Gulf of Mexico Energy Security Act (GOMESA) and these revenues provide a critical source of funding for coastal restoration projects in Louisiana. A similar structure to allocate revenues from wind energy projects in the Gulf of Mexico has yet to be established, though pending legislation may expand the limits of GOMESA to include wind revenues. An in-depth research report from the Blanco Center titled, "Offshore Wind in the Gulf of Mexico: Natural Resource Revenue Potential" examines the potential for wind development and estimates potential

HIGHLIGHTS

- **Three BOEM lease areas near Louisiana and Texas have potential to generate 1,414 MW of electricity—enough for 1.3 million homes.**
- **Developer interest in U.S. offshore wind has grown in recent decades, with lease bid amounts frequently exceeding $1,000 per acre.**
- **Shallow waters and a well-established marine industry and workforce will be appealing to Gulf of Mexico wind developers, but low wind speeds, weather risk, and other conditions could increase installation costs.**
- **Bonus bids from wind lease auctions in the Gulf of Mexico could generate an estimated $400 million in federal revenue, but there is a great deal of uncertainty about these revenues. Other revenues like rents and operating fees are likely to generate relatively small amounts of revenue compared to bonus bids.**
- **The allocation of federal wind revenues is currently uncertain. Pending legislation in the U.S. Congress may expand existing laws like GOMESA to allocate federal wind revenues to state and local governments.**

revenues from federal wind leases in the Gulf of Mexico. This policy brief offers a summary of the report, with an overview of the growth potential for wind in the Gulf of Mexico and estimates of revenue from offshore wind lease bids, rents, and operating fees.

WIND ENERGY IN THE GULF OF MEXICO

The Gulf of Mexico has direct access to abundant offshore wind resources. The Department of Interior (DOI)'s Bureau of Ocean Energy Management (BOEM) is currently offering lease areas with potential to generate 1,414 MW (1.4 GW) of electricity. This represents a small but important first step for wind development in light of the National Renewable Energy Laboratory's 2022 assessment that the Gulf of Mexico has a total offshore wind energy technical capacity potential in of 696 GW from fixed-bottom turbines and 867 GW from floating-bottom, for a total of 1563 GW.3 These figures provide a technical ceiling for wind generation in the Gulf of Mexico, however with no precedent to look to, there is a degree of uncertainty about the level of interest among potential developers in wind lease sales in the near future.

Compared with offshore wind areas in other parts of the U.S., the Gulf of Mexico appeals to developers in certain ways but not all. First and foremost, the Gulf of Mexico has fundamental infrastructure for energy transport in place thanks to the region's system of pipelines and specialization in offshore oil and gas production. A well-established pipeline system could lower costs associated with power generation, including transmitting green hydrogen produced from offshore wind.4 An existing workforce with experience from oil and gas related industries including offshore natural resource development could lower labor costs for offshore wind project construction and operation.5 Some features of the Gulf of Mexico's natural environment (e.g., shallow water depths, typically mild weather, and calm sea states) are favorable to the development of offshore wind production. In contrast several other environmental aspects of the Gulf of Mexico (e.g., lower annual average wind speeds, softer sea bottom substrate, and extreme weather events such as tropical storms and hurricanes) make development of offshore wind more challenging in the Gulf of Mexico.6 Other challenges for developers include low average electricity prices in the region and a lack of renewable portfolio standards in all Gulf of Mexico states except Texas, which has already achieved its goal. These factors could lead to relatively low revenues from electricity produced from offshore wind energy.

In October 2022, BOEM designated two wind energy areas totaling 682,540 acres in the Gulf of Mexico (Figure 1), including a 508,265-acre area whose geographic center is approximately 35 nautical miles from the coastline of Texas, and a 174,275-acre area whose geographic center is 38 nautical miles from the coastline of Louisiana.7 In February 2023, the DOI announced a Proposed Sale Notice for three lease areas in the Gulf of Mexico that could be sufficient to power over 1.3 million homes once fully operational. These lease areas include (1) a 102,480-acre area near Lake Charles, Louisiana; (2) a 102,480-arce area near Galveston, Texas; and (3) a 96,786-arce area also offshore of Galveston.8 As of May 2023, BOEM has sought public comments on these proposed areas for offshore wind leasing.9

The Gulf of Mexico designated wind energy areas are outside of state waters and under federal jurisdiction. While states have options for some non-oil and gas revenue sharing in the Outer Continental Shelf Lands Act 8(g) zone, these rules would not apply to the proposed lease areas offered by BOEM because they are further offshore and outside of the 8(g) zone.10

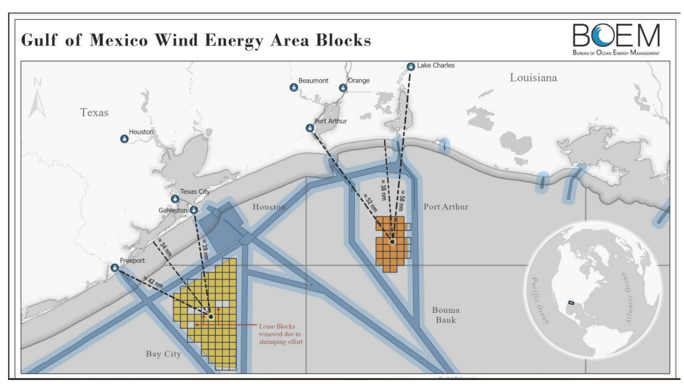

Figure 1. Finalized wind energy area blocks in the Gulf of Mexico, announced by BOEM in February of 2023.

Public revenues from offshore wind production are generated from four sources in different phases of leasing processes. BOEM is in charge of overseeing the leasing and permitting of federal offshore wind projects on the U.S. Outer Continental Shelf, and

DOI's Office of Natural Resources Revenue (ONRR) is authorized to collect four categories of revenues during different stages.[11]

1. Developers secure leases from BOEM. For competitive lease sales, ONRR collects a **bonus bid** from the highest bidder in exchange for granting the lease.
2. Before energy production begins, the developer pays an **annual rent** to ONRR. Annual rent payments are based on the area granted in the lease, using a set rate at $3 per acre.
3. Once operations begin, ONRR collects **operating fees** from the developer, similar to royalties in oil and gas production. The operating fee is set to 2% of the anticipated revenue of wind energy produced in the facility.
4. Additionally, the ONRR also collects other revenues such as settlement agreements and interest payments.

This framework is similar to that for oil and gas production in federal waters but varies slightly regarding rents and operating fees: oil and gas companies pay rents for the entire lease area until production begins anywhere within the leased block, yet wind companies would pay rents for the portion of a lease area not under operation while paying operating fees for areas under production as a development is gradually built out.

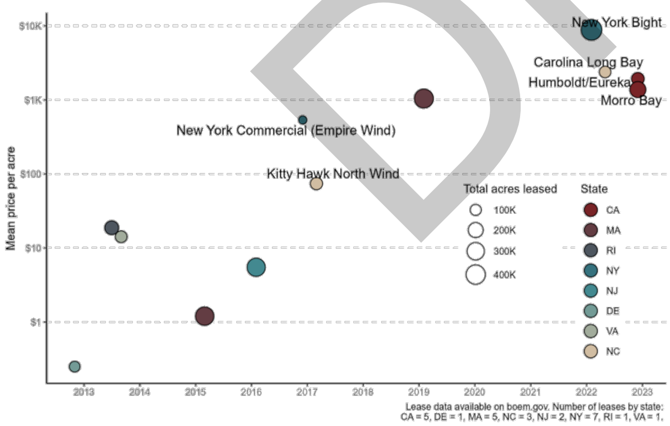

Figure 2. Mean bonus bid per acre for offshore wind lease auctions in the United States from 2012 to 2022. Note the logarithmic scale.

While the unit price of rent and operating fees have been similar across most lease sales to date, records of offshore wind leasing demonstrates a fluctuating increase trend in bonus bids, which is particularly clear after controlling for the acreage of each lease sale (Figure 2).12 Early lease sales attracted lower levels of bidding while sales in more recent years reached record numbers. In 2022, the first lease areas in the Pacific were sold in California with bonus bids of $1,400 per acre (total cash bids at $332 million, Morro Bay) and $2,000 per acre (total cash bid at $255 million, Humboldt/Eureka), respectively. In the Atlantic, the Carolina Long Bay lease sold for a cash bid similar to those off the California coast (total cash bid at $262.5 million), with a higher price-per acre at $2,400. The sale of the New York Bight made headlines with record-breaking bonus bids of $8,831 per acre, and a total cash bid of $4.37 billion.

Bonus bids for the Lake Charles and Galveston leases are expected to be the first large revenue sources for the offshore wind leases in Gulf of Mexico. They represent a significant revenue opportunity, yet the level of revenue remains uncertain.

GULF OF MEXICO WIND REVENUE SCENARIOS

Of the primary wind lease revenue sources, bonus bids are the most difficult to estimate for the Gulf of Mexico because developer bids are dependent on a number of factors, many of which are unique to the region, making it difficult to assign the value of other lease areas as a comparison. As mentioned in the previous section, the Gulf of Mexico differs from other wind leasing areas in the U.S. Relatively low wind speeds will require taller and larger turbines, and a soft sea floor means more costly platforms will be required to anchor the turbines in place. In addition, Louisiana and Texas both have lower than average electricity prices compared to other offshore wind areas that have already been leased. Texas is the only state in the region with a renewable portfolio standard, but those standards have been met leaving minimal additional regulatory pressure on the market to secure new renewable resources. These factors could serve to lower developer interest in Gulf of Mexico lease areas relative to other areas.

In comparison to recent lease auctions in California and North Carolina, the Gulf of Mexico region has some advantages. For instance, Louisiana and Texas both have strong maritime and manufacturing industries with decades of experience installing platforms in the Gulf of Mexico. In fact, Gulf Island

Fabrication, a Louisiana-based company, was selected to build the turbine platforms for the first offshore U.S. wind farm at Block Island in Rhode Island.[13] Even with a soft sea floor, the Gulf has shallow waters and normally calm seas that will be attractive to wind developers in contrast to the deeper and rougher waters along the coast of California, which will require expensive and difficult to install floating platforms. Despite the engineering challenges facing offshore wind developers in California, the three 2022 lease sales in the region fetched bonus bids ranging from $1,400-$2,400 per acre and, as seen in the New York Bight lease sale, the potential for lease bids could reach even higher. In fact, as seen in Figure 2, per acre lease amounts for offshore wind have been increased over time, indicating a growing level of interest from developers. This relates to another factor that could have a significant impact on final bid amounts: the level of competition among bidders. It is currently unknown how many developers plan to bid in the upcoming Gulf of Mexico lease auction, but it is possible that bidder competition was a factor in the record bid amount paid for the NY Bight lease in February of 2023, which saw 18 developers bidding in 64 rounds.

Table 1. Gulf of Mexico wind lease revenue scenarios.

	Low Scenario	Medium Scenario	High Scenario
Bonus Bids (2023)	$78.7 M	$404.6 M	$2,664 M
Annual Rents (2023)	$0.2 M	$0.7 M	$0.9 M
Annual Operating Fees (2030)	$1.2 M	$3.2 M	$13.2 M
2023 Leased Capacity (MW)	707	1,061	1,414
2030 Operational (MW)	707	1,414	2,829

The revenue estimates outlined in Table 1 reflect the considerations mentioned above and others discussed in greater detail in the main report. The more plausible Medium Scenario reflects an average bonus bid of $2,384 per acre (an equivalent value to the Carolina Long Bay sale of May 2022), with 75% of the leasable area, or 226,000 acres sold in 2023. Based on this scenario, the expected bonus bid amount is just over $400 million dollars. Annual rents would be nearly $700,000 per year, until the lease is fully operational by 2030 (in addition to an assumption that the original 25% of unleased areas being reauctioned and leased), after which time the developers would pay estimated operating fees of $3.7 million per year.

The Low Scenario contemplates a lower average bid price of $1,043 per acre (equal to a recent Massachusetts area auction amount) and only 50% of the leasing capacity auctioned and operational by 2030. Finally, the High Scenario shows the revenue potential if the Gulf of Mexico lease areas were to receive bids equal to the record $8,831 per acre seen in the NY Bight lease auction. In the High Scenario, 100% of the current lease areas would be auctioned, followed by a subsequent lease issuance that would double of the lease area size and result in 2,829 MW of operational capacity by 2030.

CONCLUSION

Wind offers an exciting opportunity for developing new energy in the Gulf of Mexico. The three BOEM lease areas off the coast of Texas and Louisiana will offer the opportunity to create wind energy that could power over a million homes or support industry. Previous wind leases offer an opportunity to estimate potential revenues from wind but must be framed within the unique differences and challenges that wind energy development faces in the Gulf of Mexico region. While bid amounts for offshore wind leases have trended upward in recent years, the low wind speeds and occasionally challenging weather conditions in the Gulf Mexico may reduce the value in this region. Lack of renewable energy portfolio standards outside of Texas and low average electricity prices across the Gulf of Mexico also present challenges. Nevertheless, the robust offshore workforce, shallower waters, and established marine industry (especially offshore transportation, construction, and manufacturing industries) provide valuable support for development of Gulf of Mexico wind projects.

The changing landscape for revenue sharing as it pertains to wind in the Gulf of Mexico remains a concern for state and local governments. Bonus bids provide the largest opportunity for revenue from wind; annual rents and operating fees will provide a smaller long-term revenue stream. While current revenue sharing for oil and gas projects provides substantial annual revenue for state and local governments, wind energy does not generally have the same authorization to share revenue with state and local government. Pending legislation at the federal level may update or expand existing revenue sharing agreements for oil and gas so that they also include other energy sources. However, no enacted federal legislation currently provides revenue sharing opportunities for local and state governments in the three proposed wind energy area blocks in the Gulf of Mexico.

REFERENCES

1. Gulf of Mexico Fact Sheet. *U.S. Energy Information Administration* https://www.eia.gov/special/gulf_of_mexico/ (2022).
2. The White House. *FACT SHEET: President Biden Celebrates New Commitments toward Equitable Workforce Development for Infrastructure Jobs*. https://www.whitehouse.gov/briefing-room/statements-releases/2022/11/02/fact-sheet-president-biden-celebrates-new-commitments-toward-equitable-workforce-development-for-infrastructure-jobs/ (n.d.).
3. Lopez, A. *et al.* Offshore Wind Energy Technical Potential for the Contiguous United States. https://www.nrel.gov/docs/fy22osti/83650.pdf (2022).
4. Thorson, J. *et al.* Unlocking the Potential of Marine Energy Using Hydrogen Generation Technologies. https://www.nrel.gov/docs/fy22osti/82538.pdf (2022).
5. Stefek, J. *et al.* U.S. Offshore Wind Workforce Assessment. https://www.nrel.gov/docs/fy23osti/81798.pdf (2022).
6. Musial, W. *et al.* Offshore Wind in the US Gulf of Mexico: Regional Economic Modeling and Site- Specific Analyses. (2020).
7. John, F. BOEM Designates Two Wind Energy Areas in Gulf of Mexico | Bureau of Ocean Energy Management. https://www.boem.gov/newsroom/press-releases/boem-designates-two-wind-energy-areas-gulf-mexico (2022).
8. Department of the Interior. Proposed Sale Notice for Commercial Leasing for Wind Power Development on the Outer Continental Shelf in the Gulf of Mexico (GOMW–1). https://www.boem.gov/sites/default/files/documents/renewable-energy/state-activities/BOEM_FRDOC_0001-0629.pdf (2022).
9. U.S. Department of the Interior. Interior Department Proposes First-Ever Offshore Wind Sale in Gulf of Mexico. https://www.doi.gov/pressreleases/interior-department-proposes-first-ever-offshore-wind-sale-gulf-mexico (2023).
10. NOAA Office for Coastal Management. So What? Zone, Limit, and Maritime Jurisdictions in the Marine Environment. https://www.marinecadastre.gov/SiteCollectionDocuments/SoWhat_MarineBoundaries_final_template.pdf.
11. U.S. Department of the Interior. Offshore Renewables | How revenue works. https://revenuedata.doi.gov/how-revenue-works/offshore-renewables/.
12. Bureau of Ocean Energy Management. Lease and Grant Information | Bureau of Ocean Energy Management. https://www.boem.gov/renewable-energy/lease-and-grant-information.
13. Larino, J. Gulf Island Fabrication wins foundation work for first U.S. offshore wind farm. *The Times-Picayune* (2014).
14. Ionna22. *Block Island offshore wind farm*. (2017).